

7(a)	volt / ampere	B1
7(b)	$R = \rho L / A$	C1
	$A = 460 \times 10^{-9} \times 2.5 / 3.2$	C1
	$= 3.6 \times 10^{-7} \text{ m}^2$	A1
7(c)(i)	energy is dissipated in the internal resistance/ r	B1
7(c)(ii)	$E = IR + Ir$ or $E = I (R + r)$	B1
7(c)(iii)	$P = I^2 R$ or $P = I^2 r$	C1
	$I = E / 2r$	A1
	(so) $P = E^2 / 4r$	